

SMARTEDIT: A HUMAN-IN-THE-LOOP AI FRAMEWORK FOR MULTI-FOOTAGE VIDEO EDITING

SUDHI BABU
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Mini Project

Guided By:
[Guide Name]

Department of Computer Applications
Government Engineering College, Thrissur, Kerala

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Introduction

- ▶ Video editing is a time-consuming process requiring editors to manually review footage, remove unwanted segments, arrange clips, and construct a coherent timeline.
- ▶ Modern AI-powered editing tools automate parts of this workflow, but often offer limited user control and can produce edits that do not match the creator's intended style.
- ▶ This creates a need for an intelligent editing system that combines the speed of AI with the creativity and decision-making of human editors.

Introduction

- ▶ Creating the initial rough cut – identifying useful footage, removing shaky clips, trimming silence, and arranging clips – is repetitive yet consumes a large share of total editing time.
- ▶ Fully automated AI editors generate videos quickly but fail to understand the editor's creative intention, storytelling requirements, or personal style.
- ▶ Human-in-the-Loop (HITL) AI combines the speed of automation with human decision-making, letting the editor review, accept, reject, or modify AI suggestions before finalizing the output.

Objectives:

- ▶ To design and develop SmartEdit, an AI-assisted Human-in-the-Loop video editing system that interprets natural language editing prompts, analyzes video and audio content, and generates a reviewable rough cut – accelerating rough-cut creation while keeping the editor in complete creative control.

Tilekbay, T. et al., “ExpressEdit: A Multimodal System for Video Editing with Natural Language and Sketching,” *Proc. ACM CHI Conference on Human Factors in Computing Systems*, 2024.

- ▶ Enables users to edit videos through natural language prompts combined with sketch-based interactions, improving accessibility and user interaction.
- ▶ **Observation:** Supports editing commands but provides no AI-assisted rough-cut generation, computer vision analysis, or Human-in-the-Loop review – the gap SmartEdit targets.

Chen, Z. et al., “EditBoard: A Comprehensive Evaluation Benchmark for Text-based Video Editing,” *arXiv preprint arXiv:2402.09257*, 2025.

- ▶ Introduces a comprehensive benchmark for evaluating text-based AI video editing models and comparing their performance.
- ▶ **Observation:** Useful for evaluation, but offers no interactive editing system – confirming the need for a working, user-facing HITL application rather than a benchmark alone.

Wang, K. et al., “LAVE: An LLM-powered Agent for Video Editing,” *arXiv preprint arXiv:2402.19131*, 2024.

- ▶ Uses Large Language Models to understand user instructions and automate video editing tasks.
- ▶ **Observation:** Automates editing well but gives limited user control during editing and lacks Human-in-the-Loop refinement – motivating SmartEdit’s review-before-export workflow.

Shekar, K. et al., “AI-Based Intelligent Video Editing System,” *Proc. Int. Conf. Advancements in Smart, Secure and Intelligent Computing (ASSIC)*, 2025.

- ▶ Proposes an AI-assisted framework for automatic scene selection and video editing.
- ▶ **Observation:** Does not integrate natural language prompt interpretation, interactive timeline editing, or user-guided refinement – all core to SmartEdit’s design.

Tu, S. et al., “Video Highlight Segment Recognition and Automatic Editing via Computer Vision,” *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2025.

- ▶ Presents an advanced computer vision framework focused on automated understanding of video content and editing operations.
- ▶ **Observation:** Emphasizes full automation with no Human-in-the-Loop editing, natural language prompts, or personalized workflow – reinforcing SmartEdit’s collaborative approach.

Li, B. et al., “VEU-Bench: Towards Comprehensive Understanding of Video Editing,” *CVPR*, 2025.

Biyyala, V. et al., “SST-EM: Advanced Metrics for Evaluating Semantic, Spatial and Temporal Aspects in Video Editing,” *arXiv preprint*, 2025.

- ▶ Both works advance the *evaluation* of AI-generated edits – benchmarking video-editing understanding and proposing richer semantic/spatial/temporal quality metrics.
- ▶ **Observation:** Strong evaluation tooling exists, but not paired with an interactive system that lets a human editor act on those judgments in real time – SmartEdit closes this loop directly in the editing workflow.

Literature Review

Reference	Approach	Brief Observation
[1] Tilekbay et al. (2024)	NL + sketch video editing	No AI rough-cut generation or HITL review.
[2] Chen et al. (2025)	Text-based editing benchmark	Evaluation only, not an interactive system.
[3] Wang et al. (2024)	LLM video editing agent	Automates editing with limited user control.
[4] Shekar et al. (2025)	AI scene selection & editing	No NL prompts or interactive refinement.
[5] Tu et al. (2025)	CV-based automated editing	Fully automated; no human review step.
[6],[7] Li et al. / Biyyala et al. (2025)	Evaluation benchmarks & metrics	Assess edit quality but don't drive an interactive tool.

- ▶ **Adobe Premiere Pro:** Comprehensive timeline editing with AI tools such as Auto Reframe and Speech Enhancement, but rough-cut creation and refinement of AI suggestions remain manual.
- ▶ **DaVinci Resolve:** Neural Engine adds object detection, facial recognition, and automatic masking, but focuses on post-production rather than natural-language, user-guided rough cuts.
- ▶ **Runway ML:** Cloud-based AI editing (background removal, object tracking, video generation) with complete automation and limited user control over internal decisions.
- ▶ **Common limitations:** time-consuming manual editing; limited human control in AI-based tools; no natural language editing support; isolated, non-integrated AI features; limited transparency/explainability of AI decisions.

Proposed System

- ▶ SmartEdit: an AI-assisted Human-in-the-Loop video editing application built with Qt (C++), Python, OpenCV, Librosa, and FFmpeg.
- ▶ **Prompt Interpretation Module:** A Small Language Model (SLM) converts natural language editing instructions into structured editing commands.
- ▶ **Video Analysis Module:** OpenCV detects shaky footage, blurred scenes, and duplicate/low-quality clips.
- ▶ **Audio Analysis Module:** Librosa detects silent regions and low-energy segments for automatic trimming.
- ▶ **AI-Assisted Rough Cut Generation:** Combines prompt, video, and audio analysis into a preliminary timeline matching the requested style and duration.
- ▶ **Human-in-the-Loop Review Module:** Users preview, accept, reject, and rearrange AI suggestions before anything is finalized.
- ▶ **Video Rendering and Export Module:** FFmpeg encodes and exports the finalized timeline at high quality.

Requirements Specification

Hardware Requirements

- ▶ **Processor:** Intel Core i5 (11th Gen or higher) / AMD Ryzen 5 or higher
- ▶ **Memory:** Minimum 8 GB RAM (16 GB recommended)
- ▶ **Storage:** Minimum 100 GB SSD free space
- ▶ **Graphics:** NVIDIA GeForce GTX 1650 or higher (recommended)
- ▶ **Display:** Full HD (1920×1080) monitor

Requirements Specification

Software Requirements

- ▶ **Operating System:** Windows 10/11 (64-bit); adaptable to Linux with Qt support
- ▶ **Languages:** C++17 or later (desktop app), Python 3.11 or later (AI modules)
- ▶ **GUI Framework:** Qt 6 – interface, timeline editor, media playback
- ▶ **Computer Vision:** OpenCV 4.x – shaky footage/frame-quality detection
- ▶ **Audio Processing:** Librosa – silence detection, feature extraction
- ▶ **AI Model:** Small Language Model (SLM) – natural language prompt interpretation
- ▶ **Multimedia Engine:** FFmpeg – decoding, trimming, encoding, export

- ▶ **Dev Tools:** Qt Creator, Visual Studio Code, Git, Python virtual environments

Requirements Specification

Functional Requirements

- ▶ **Video Import & Project Creation:** Import multiple video files; validate format; extract metadata (duration, resolution, frame rate, audio).
- ▶ **Prompt-Based Editing:** Users specify style, target duration, clip arrangement, and preferences (e.g. remove silence/shaky footage) in natural language; SLM converts these into structured instructions.
- ▶ **Video Analysis:** Detect shaky footage, blurred/low-quality frames, and duplicate clips via OpenCV.
- ▶ **Audio Analysis:** Detect silence, low-energy segments, and speech activity via Librosa.
- ▶ **AI-Assisted Rough Cut Generation:** Arrange clips per prompt, remove unnecessary/silent footage, and match target duration.

- ▶ **Human-in-the-Loop Review:** Preview, accept, reject, or re-generate the rough cut – no edit is applied without approval.
- ▶ **Manual Timeline Editing:** Trim, rearrange, insert, or remove clips after the rough cut is approved.
- ▶ **Rendering & Export:** FFmpeg-based export with progress tracking and quality/sync preservation.

Non-Functional Requirements

- ▶ **Performance:** Responsive playback and low-latency AI analysis even for high-resolution video.
- ▶ **Usability:** Clean, intuitive interface suited to both beginners and professional editors.
- ▶ **Accuracy & Reliability:** Consistent detection and suggestions; no permanent edit without user approval.
- ▶ **Security & Data Privacy:** Project files accessible only to authorized users; no unconfirmed modification/deletion.
- ▶ **Maintainability & Extensibility:** Modular architecture separating UI, AI modules, and rendering engine.
- ▶ **Scalability:** Supports multiple video files of varying resolution/duration within one project.

Input and Output

- ▶ **Input:** One or more raw video files imported from local storage, plus a natural language prompt describing editing style and target duration.
- ▶ **Input Format:** Standard multimedia video files supported by OpenCV/FFmpeg.
- ▶ **Output:** An AI-generated, reviewable rough-cut timeline followed by a final rendered video reflecting user-approved and manually refined edits.
- ▶ **Output Format:** Exported video file in a supported format, produced via FFmpeg with preserved video/audio quality and synchronization.

Feasibility Study

- ▶ **Technical Feasibility:** Built on mature, well-supported technologies (Qt, Python, OpenCV, Librosa, FFmpeg, SLMs); runs on standard desktop hardware without specialized infrastructure.
- ▶ **Operational Feasibility:** Fits naturally into existing workflows of content creators, YouTubers, wedding videographers, and news editors by automating rough cuts while preserving full human control.
- ▶ **Economical Feasibility:** Relies primarily on open-source tools (Qt, Python, OpenCV, Librosa, FFmpeg, Git), keeping development and deployment costs low.

Conclusion

- ▶ SmartEdit combines natural language prompt interpretation, computer vision, and audio signal processing within a single Human-in-the-Loop desktop application.
- ▶ Unlike existing fully automated or purely evaluative tools, it focuses on generating a reviewable rough cut that the editor can accept, reject, or refine before export.
- ▶ Aims to significantly reduce rough-cut creation time while keeping complete creative control with the human editor.
- ▶ Built on mature open-source technologies, making it practical and affordable for content creators and media professionals.

References

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- [7] V. Biyyala, B. C. Kathuria, J. Li and Y. Zhang, “SST-EM: Advanced Metrics for Evaluating Semantic, Spatial and Temporal Aspects in Video Editing,” *arXiv preprint*, 2025.

Thank You